



INTERNATIONAL CONFERENCE

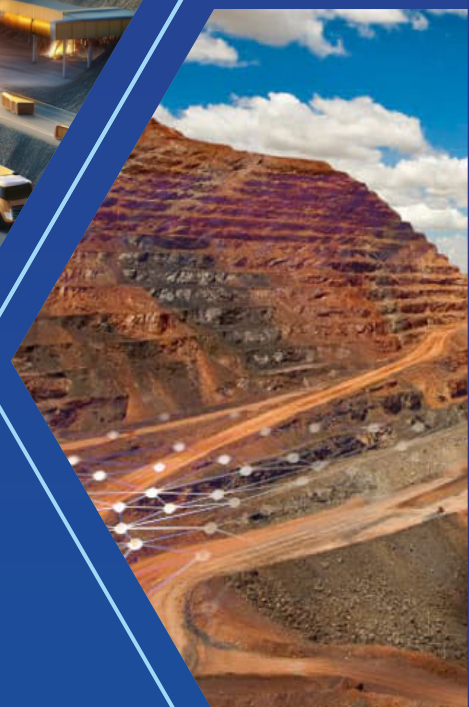
“Digital Intelligence for Green Mining and Industrial Networks” (DIGMIN-2025)



GJLT, IIT-ISM, Dhanbad



12-13 September, 2025



Organized by

Department of Mining Engineering

INDIAN INSTITUTE OF TECHNOLOGY (ISM) DHANBAD – 826004
JHARKHAND, INDIA

CONFERENCE CONCEPT

Digital Intelligence for Green Mining and Industrial Networks (DIGMIN) – 2025 will serve as India's flagship conference at the intersection of digitalization, intelligent automation, and sustainable mining practices. DIGMIN-2025 is envisioned as a premier conference focused on accelerating the digital transformation of the mining sector. Anchored on the pillars of Digitalization, Intelligent Systems, Green Technologies, Mining 5.0, Industrial Integration, and Next-generation Infrastructure, the conference aims to bring together mining professionals, regulatory personnel, academic researchers, technologists, sustainability experts, policymakers, and industry leaders to deliberate on the accelerated adoption of next-generation digital technologies in mining and allied industrial ecosystems. DIGMIN-2025 will showcase AI/ML-driven risk prediction advancements, robotics, IoT-enabled automation, and eco-efficient mining practices. The event will foster collaboration between public institutions, private players, and startups through panel discussions, technology exhibitions, and policy roundtables. The goal is to unlock hidden opportunities in mining through smart digital systems, drive sustainable resource management, and build a roadmap for a resilient, technology-integrated, and environmentally responsible mining ecosystem in India. It will be a landmark conference organized as part of the Centenary Celebrations of IIT (ISM) Dhanbad, commemorating 100 years of excellence in mining education and research.

ABOUT IIT (ISM) DHANBAD



The Indian Institute of Technology (Indian School of Mines) Dhanbad is situated in India's prime coking coal belt, which is 260 km from Kolkata city. IIT (ISM) Dhanbad offers all the amenities of a world-class academic institution and is fully residential. The Indian School of Mines was officially inaugurated on December 9th, 1926, by Lord Irwin, the then Viceroy of India, to meet the demand for trained manpower associated with mining activities in the country, focusing on Mining and Applied Geology disciplines. The Indian School of Mines was granted the status of Indian Institute of Technology (IIT) on September 06, 2016, by the Government of India, and it was renamed the Indian Institute of Technology (Indian School of Mines) Dhanbad. Since its inception, the institute has significantly broadened its scope of activities, evolving into a comprehensive technology education institute. IIT (ISM) offers B.Tech. (4 Years), Integrated M. Tech. (5 Years), Master of Science/Arts and Technology, Master of Business Administration, and Ph.D. programs. IIT(ISM) Dhanbad has made significant contributions to the advancement of mining, minerals, petroleum, and groundwater exploration in India.

ABOUT THE DEPARTMENT



The institute/ deemed university, Indian School of Mines (ISM), Dhanbad, was established in 1926, with the Department of Mining Engineering founded by the then Viceroy of India, Lord Irwin. The school came under the purview of the Indian Institute of Technology in 2016 and was renamed the Indian Institute of Technology (Indian School of Mines), IIT(ISM). Over the past 99 years, it has developed and grown to become the country's

largest department of its kind, boasting excellent teaching and research facilities alongside widespread activities in the coal, metalliferous mining, and construction sectors. In the QS World University Rankings by Subject 2025, IIT (ISM) Dhanbad is ranked 20th globally for Mineral and Mining Engineering, making it the top-ranked Indian institution in this subject area.

THEMES OF THE CONFERENCE

Digital Foundations for Smart Mining

- Integration of IoT, cloud, and edge computing in mine operations

Robotics and Automation in Harsh Mining Environments

- Autonomous drilling, blasting, and hauling systems
- Swarm robotics for exploration and maintenance
- Human-robot collaboration in underground and confined spaces

Edge AI and Real-Time Analytics in Mining

- Low-latency AI applications for critical mine operations
- Edge devices for environmental and structural monitoring
- Federated learning in distributed mine networks

Geospatial Intelligence and Digital Mapping

- AI-powered remote sensing and satellite imaging
- 3D subsurface modeling using LiDAR and hyperspectral data
- Integration of GIS, drones, and ground-penetrating radar

Digital Resilience and Disaster Management in Mining

- Design of intelligent operation centers
- Integration of AR/VR for remote inspections and diagnostics, enabling remote decision-making with digital twins

Energy Efficiency and Process Optimization

- AI for optimizing crushing, grinding, and material handling
- Dynamic control of furnaces and kilns using digital feedback
- CPS-enabled real-time energy monitoring and reduction strategies

CALL FOR PAPERS

The technical papers on the Conference theme are invited for oral presentation. The received papers will be peer-reviewed. It is intended to publish the proceedings of the Conference. Contributors must submit their original technical documents in the desired format in MS Word through email attachment on or before 15 August 2025.

Official Email ID for all correspondences and sending papers: sagarwal@iitism.ac.in

Format for preparing abstracts: **Title:** Times New Roman 12 bold, **Author:** Times New Roman 10.5, **Affiliation:** Times New Roman 10 italic, **Headings:** Times New Roman 10 bold, **Sub-Headings:** Times New Roman 12 bold, **Paragraph texts:** Times New Roman 10. **Figures:** Figure 1, Figure 2, etc. **Tables:** Table 1, Table 2, etc. Figures and tables should be placed at their references within the text. References should be cited in Chicago style only in the main text and arranged alphabetically, with heading references.

SPONSORSHIP

Diamond Sponsor	INR 10,00,000/- (10 delegates free of cost)
Gold Sponsor	INR 7,00,000/- (7 delegates free of cost)
Silver Sponsor	INR 5,00,000/- (5 delegates free of cost)
Bronze Sponsor	INR 3,00,000/- (3 delegates free of cost)

(excluding GST@18%)

REGISTRATION FEE

Registration fee (excluding GST@18%)	Indian delegates, INR	Foreign delegates, USD
Delegates (Academics & Research Organization)	7,000	150
Students/Research Scholars	3,000	50
Delegates (Industry/ Govt. Agency)	10,000	200

ADVERTISEMENT IN THE SOUVENIR

Back cover	INR 30,000/- (color)
Front inside	INR 25,000/- (color)
Back inside	INR 25,000/- (color)
Full page (other)	INR 15, 000/-(color)

(excluding GST@18%)

EXHIBITION

Approximately 10 exclusive stalls, each of 9.0 sq. m. (3.0 m x 3.0 m) with electricity connections, will be made available @INR 1,50,000.00 (USD 2000 for overseas exhibitors) per stall (excluding GST @18%) to provide an unparalleled opportunity to the exhibitors to showcase their products and software related to the subject matter. These stalls will be provided on a 'first come, first served' basis. The stall charges include the registration fee of two representative of the company.

The Exhibition will be opened to visitors from 9:30 AM to 6:00 PM during 12-13 September 2025, ensuring ample time to engage with a diverse audience. Exhibitors are requested to contact the Conference Secretariat to book the stalls.

BANK DETAILS FOR PAYMENT

Name: IIT(ISM) SPECIAL FUND	Bank: Canara Bank; Saraidhela Branch, Dhanbad
A/C No.: 0986101024892	IFSC: CNRB0000986;
GSTIN: 20AAAAI0686D1ZA	MICR code: 826015003;

KEY DETAILS

Call for abstracts	01.06.2025
Receipt of abstracts	30.06.2025
Review of abstracts and decision notification	15.07.2025
Receipt of full papers	15.08.2025
Review of full papers and decision intimation	21.08.2025
Intimation of acceptance of papers	31.08.2025

CORRESPONDENCE

Prof. Anindya Sinha

Convener and Chairman

Department of Mining Engineering,
Indian Institute of Technology
(Indian School of Mines), Dhanbad – 826004
Mobile No.: 7766908001
Email: anindya@iitism.ac.in

Prof. Siddhartha Agarwal

Co-Convener

Department of Mining Engineering,
Indian Institute of Technology
(Indian School of Mines), Dhanbad – 826004
Mobile No.: 9335536412
Email: sagarwal@iitism.ac.in

Prof. Ajeet Yadav

Treasurer

Department of Mining Engineering,
Indian Institute of Technology
(Indian School of Mines), Dhanbad – 826004
Mobile No.: 7607222169
Email: ajeet@iitism.ac.in

Mr. Rajul Dwivedi & Ms. Pratibha Sharma

Student Co-Ordinators

Department of Mining Engineering,
Indian Institute of Technology
(Indian School of Mines), Dhanbad – 826004
Mobile No.: 7772969347 & 6372685665
Email: 23dp0082@iitism.ac.in &
23dr0280@iitism.ac.in

ORGANIZING COMMITTEE

Chief Patron

- Prof. Prem Vrat
Chairperson, BoG, IIT(ISM) Dhanbad

Patron

- Prof. Sukumar Mishra
Director, IIT(ISM) Dhanbad

Co-Patron

- Prof. Dheeraj Kumar
Dy. Director, IIT(ISM) Dhanbad

Convener and Chairman

- Prof. Anindya Sinha

Co-Convener

- Prof. Siddhartha Agarwal

Treasurer

- Prof. Ajeet Yadav

Co-Ordinators

- Ms. Pratibha Sharma
Mr. Rajul Dwivedi

International Advisory Committee

- Prof. YP Chugh
USA
- Mr. Sietse van der Woude
SA
- Dr. Sekhar Bhattacharyya
USA
- Dr Sihe Nhleko
SA
- Prof. Snehamoy Chatterjee
USA
- Prof. Rajive Ganguli
USA
- Prof. Cihan Dagli
USA
- Prof. Erkan Topal
Australia
- Prof. Hooman Askari-Nasab
Canada
- Prof. Simit Raval
Australia
- Prof. Ashish Kumar
USA
- Prof. Marek Borowski
Poland
- Rudrajit Mitra
USA
- H. Sebnem Düzgün
USA
- D. R. Nagaraj
USA
- Sampurna Arya
USA
- Victor Tenorio
USA

NATIONAL ADVISORY COMMITTEE

- Prof. Dheeraj Kumar
Deputy Director, IIT (ISM)
- Prof. M. K. Singh
Dean (Academic), IIT (ISM)
- Prof. Sarat Kr Das
Dean (Faculty), IIT (ISM)
- Prof. RM Bhattacharjee
Dean (IRAA), IIT (ISM)
- Prof. Sagar Pal
Dean (R&D), IIT (ISM)
- Prof. SK Gupta
Dean (Students Welf), IIT (ISM)
- Prof. AR Dixit
Dean (Infrastructure), IIT (ISM)
- Prof. R Singh
Dean (Corp. Comm.), IIT(ISM)
- Prof. Alok Kr Das
Dean (IIE), IIT (ISM)
- Shri Prabodh Pandey
Registrar, IIT (ISM)
- Prof. Biswajit Samanta
HOD/MIN, IIT KGP
- Prof. S. Gupta
HOD/MIN, IIT (BHU) Varanasi
- Prof. NC Dey
HOD/MIN, IEST Shibpur
- Prof. A. K. Gorai
HOD/MIN, NIT Rourkela
- Prof. Harsha Vardhan
HOD/MIN, NIT Surathkal
- Prof. B. R. Parate
NIT Raipur
- Prof. Ritesh Lokhande
HOD/MIN, VNIT Nagpur
- Shri Ujjal Tah
DG, DGMS Dhanbad
- Shri P. M. Prasad
CMD, CIL
- Shri A. Ghatak
Director (Technical), CIL
- Shri Manoj Kumar
CMD, CMPDI
- Shri Samiran Dutta
CMD, BCCL
- Shri J. P. Dwivedi
CMD, WCL
- Shri. Nilendu Kr. Singh
CMD, CCL
- Shri Uday A Kaole
CMD, MCL
- Shri B. Sairam
CMD, NCL
- Shri Satish Jha
CMD, ECL
- Shri Harish Duhan
CMD, SECL
- Prof. AK Mishra
Director, CIMFR Dhanbad
- Prof. VMSR. Murthy
Director, IEST Shibpur
- Shri Amitava Mukherjee
CMD, NMDC
- Shri Amarendu Prakash
CMD, SAIL
- Shri Sanjiv Kumar Singh
CMD, HCL
- Shri A. K. Saxena
CMD, MOIL
- Dr. S. Kumar Satpati
CMD, UCIL
- Shri BP Singh
CMD, NALCO

INTERNATIONAL ADVISORY COMMITTEE

- **Prof. YP Chugh**
USA
- **Mr. Sietse van der Woude**
SA
- **Dr. Sekhar Bhattacharyya**
USA
- **Dr Sihe Nhleko**
SA
- **Prof. Snehamoy Chatterjee**
USA
- **Prof. Rajive Ganguli**
USA
- **Prof. Cihan Dagli**
USA
- **Prof. Erkan Topal**
Australia
- **Prof. Hooman Askari-Nasab**
Canada
- **Prof. Simit Raval**
Australia
- **Prof. Ashish Kumar**
USA
- **Prof. Marek Borowski**
Poland
- **Rudrajit Mitra**
USA
- **H. Sebnem Düzgün**
USA
- **D. R. Nagaraj**
USA
- **Sampurna Arya**
USA
- **Victor Tenorio**
USA

NATIONAL ADVISORY COMMITTEE

- **Prof. Dheeraj Kumar**
Deputy Director, IIT (ISM)
- **Prof. M. K. Singh**
Dean (Academic), IIT (ISM)
- **Prof. Sarat Kr Das**
Dean (Faculty), IIT (ISM)
- **Prof. RM Bhattacharjee**
Dean (IRAA), IIT (ISM)
- **Prof. Sagar Pal**
Dean (R&D), IIT (ISM)
- **Prof. SK Gupta**
Dean (Students Welf), IIT (ISM)
- **Prof. AR Dixit**
Dean (Infrastructure), IIT (ISM)
- **Prof. R Singh**
Dean (Corp. Comm.), IIT(ISM)
- **Prof. Alok Kr Das**
Dean (IIE), IIT (ISM)
- **Shri Prabodh Pandey**
Registrar, IIT (ISM)
- **Prof. Biswajit Samanta**
HOD/MIN, IIT KGP
- **Prof. S. Gupta**
HOD/MIN, IIT (BHU) Varanasi
- **Prof. NC Dey**
HOD/MIN, IEST Shibpur
- **Prof. A. K. Gorai**
HOD/MIN, NIT Rourkela
- **Prof. Harsha Vardhan**
HOD/MIN, NIT Surathkal
- **Prof. B. R. Parate**
NIT Raipur
- **Prof. Ritesh Lokhande**
HOD/MIN, VNIT Nagpur
- **Shri Ujjal Tah**
DG, DGMS Dhanbad
- **Shri P. M. Prasad**
CMD, CIL
- **Shri A. Ghatak**
Director (Technical), CIL
- **Shri Manoj Kumar**
CMD, CMPDI
- **Shri Samiran Dutta**
CMD, BCCL
- **Shri J. P. Dwivedi**
CMD, WCL
- **Shri. Nilendu Kr. Singh**
CMD, CCL
- **Shri Uday A Kaole**
CMD, MCL
- **Shri B. Sairam**
CMD, NCL
- **Shri Satish Jha**
CMD, ECL
- **Shri Harish Duhan**
CMD, SECL
- **Prof. AK Mishra**
Director, CIMFR Dhanbad
- **Prof. VMSR. Murthy**
Director, IEST Shibpur
- **Shri Amitava Mukherjee**
CMD, NMDC
- **Shri Amarendu Prakash**
CMD, SAIL
- **Shri Sanjiv Kumar Singh**
CMD, HCL
- **Shri A. K. Saxena**
CMD, MOIL
- **Dr. S. Kumar Satpati**
CMD, UCIL
- **Shri BP Singh**
CMD, NALCO
- **Shri Gurdeep Singh**
CMD, NTPC
- **Shri M. Prasanna Kumar**
CMD, NLCIL
- **Shri Indra Dev Narayan**
CMD, MECL
- **Shri Surendra Kumar**
MD, OMC
- **Shri Subhrakant Panda**
MD, IMFA
- **Shri Arun Misra**
CEO & MD, HZL, Vedanta
- **Smt. Shilpa. R**
MD, HGML
- **Shri DB S. Ramam**
VP, Raw Material, TSL
- **Dr. V. Prakash**
CEO, Adani Natural Resources
- **Shri Manojit Haldar**
MD, Sandvik Mining, India

MEMBERS

- **Prof Shushanta Sarangi**
AG, IIT(ISM) Dhanbad
- **Prof. Keka Ojha**
PE, IIT(ISM) Dhanbad
- **Prof. Sravan Kumar**
FMME, IIT(ISM) Dhanbad
- **Prof. Srinivas Pasupuleti**
CE, IIT(ISM) Dhanbad
- **Prof. Sukanta Das**
EE, IIT(ISM) Dhanbad
- **Prof. Ravi Kumar Gangwar**
IIT(ISM) Dhanbad
- **Prof. Sandeep Mondal**
MS, IIT(ISM) Dhanbad
- **Prof. Aditya Kumar**
CHE, IIT(ISM) Dhanbad
- **Prof. Parthasarathi Das**
CCB, IIT(ISM) Dhanbad
- **Prof. Nirban Manna**
HSS, IIT(ISM) Dhanbad
- **Prof. Chiranjeev Kumar**
CSE, IIT(ISM) Dhanbad
- **Prof. Alok Sinha**
ESE, IIT(ISM) Dhanbad
- **Prof. S. P. Tiwari**
MNC, IIT(ISM) Dhanbad
- **Prof. B. K. Antony**
PHY, IIT(ISM) Dhanbad
- **Prof S. Chattopadhyaya**
ME, IIT(ISM) Dhanbad
- **Prof. D. P. Mishra**
MNE, IIT(ISM) Dhanbad
- **Prof. B.S. Choudhary**
MNE, IIT(ISM) Dhanbad
- **Prof. P. S. Paul**
MNE, IIT(ISM) Dhanbad
- **Prof. R.K. Sinha**
MNE, IIT(ISM) Dhanbad
- **Prof. V. G. K. Villuri**
MNE, IIT(ISM) Dhanbad
- **Prof. R. Koner**
MNE, IIT(ISM) Dhanbad
- **Prof. Patitapaban Sahu**
MNE, IIT(ISM) Dhanbad
- **Prof. G. Budi**
MNE, IIT(ISM) Dhanbad
- **Prof. Kashinath Pal**
MNE, IIT(ISM) Dhanbad
- **Prof. Md. Soyeb Alam**
MNE, IIT(ISM) Dhanbad
- **Prof. Swapnil Mishra**
MNE, IIT(ISM) Dhanbad
- **Prof. Ashok Kumar**
MNE, IIT(ISM) Dhanbad
- **Prof. D. G. Krishna**
MNE, IIT(ISM) Dhanbad
- **Prof. Bhaskara Behera**
MNE, IIT(ISM) Dhanbad
- **Prof. Ankush Galav**
MNE, IIT(ISM) Dhanbad
- **Prof. SK Behera**
MNE, IIT(ISM) Dhanbad
- **Prof. M. M. Sahoo**
MNE, IIT(ISM) Dhanbad
-



INTERNATIONAL CONFERENCE
“Digital Intelligence for Green Mining
and Industrial Networks”
(DIGMIN-2025)



GJLT, IIT-ISM, Dhanbad



12-13 September, 2025

REGISTRATION FORM

1. Name (in block letters):
2. Affiliation:
3. Address:
.....
.....
4. Mobile No.: 5. Email:
6. Participation details:
(i) Abstract submitted: Y/N
(ii) Author's Name:
(iii) Title of the Abstract:
.....
.....
7. Details of Registration Fees
Demand Draft /UTR No.: Date:
Amount:
8. Additional information, if any:
Online Link for Registration:

Sponsorships

I/We intend to sponsor/co-sponsor/nominate(No.) Delegates/ present a paper in the seminar/ advertise in the Souvenir of the DIGMIN-2025.

Date:

Signature